

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**23CSE304 Embedded Systems Laboratory**

**Activity Sheet #4**

**Activity Name:** LCD Interfacing

|                          |                         |
|--------------------------|-------------------------|
| <b>Name of Student:</b>  | <b>Agneay B Nair</b>    |
| <b>Registration No.:</b> | <b>CH.SC.U4CSE24102</b> |
| <b>Activity Date:</b>    | <b>18-08-26</b>         |
| <b>Submission Date:</b>  | <b>18-08-26</b>         |

**Assessment Rubrics:**

| <b>Component</b>  | <b>Maximum Marks</b> | <b>Marks Allotted</b> |
|-------------------|----------------------|-----------------------|
| Program           | 3                    |                       |
| Output            | 4                    |                       |
| Viva              | 2                    |                       |
| Timely Submission | 1                    |                       |
| <b>Total</b>      | <b>10</b>            |                       |

**Faculty In-charge**

**Write an embedded C program for ARM architecture to interface LCD. Debug and verify the program using IDE on a development computer system.**

**CODE:**

```
#include <LPC213x.h>

#define RS (1 << 8)
#define EN (1 << 9)

void delay(unsigned int count)
{
    unsigned int i, j;

    for(i = 0; i < count; i++)
        for(j = 0; j < 1000; j++);
}

void lcd_command(unsigned char cmd)
{
    // Clear old data from P0.16 to P0.23
    IOCLR0 = 0x00FF0000;

    // Send command to P0.16-P0.23
    IOSET0 = ((unsigned int)cmd << 16);

    // RS = 0 for command
    IOCLR0 = RS;

    // Enable pulse
    IOSET0 = EN;
    delay(1);
    IOCLR0 = EN;

    delay(2);
}

void lcd_data(unsigned char data)
{
    // Clear old data
    IOCLR0 = 0x00FF0000;

    // Send data to P0.16-P0.23
    IOSET0 = ((unsigned int)data << 16);

    // RS = 1 for data
    IOSET0 = RS;

    // Enable pulse
```

```

IOSET0 = EN;
delay(1);
IOCLR0 = EN;

delay(2);
}

void lcd_string(char *str)
{
    while(*str != '\0')
    {
        lcd_data(*str);
        str++;
    }
}

void lcd_init(void)
{
    delay(20);

    lcd_command(0x38); // 8-bit mode, 2 lines
    lcd_command(0x0C); // Display ON, cursor OFF
    lcd_command(0x06); // Auto increment cursor
    lcd_command(0x01); // Clear LCD

    delay(2);
}

int main(void)
{
    // Configure pins as GPIO
    PINSEL0 = 0x00000000;
    PINSEL1 = 0x00000000;

    // P0.8, P0.9 and P0.16-P0.23 as output
    IODIR0 = 0x00FF0300;

    lcd_init();

    // First line
    lcd_command(0x80);
    lcd_string("Hello!");

    // Second line
    lcd_command(0xC0);
    lcd_string("LPC2138 LCD");

```

```
while(1);  
}
```

## OUTPUT:

The screenshot displays the uVision IDE interface. The top menu bar includes File, Edit, View, Project, Flash, Debug, Peripherals, Tools, SVCS, Window, and Help. The toolbar contains various icons for file operations, debugging, and viewing. The main workspace is divided into several panes:

- Registers:** A list of registers (R0 to R15, CPSR, SPSR) with their current values. R0 is highlighted as the current register.
- Disassembly:** A window showing the assembly code for the current address. The instruction at address 0x000002D8 is highlighted: `76: PINSEL0 = 0x00000000; 0x000002D8 E3A00000 MOV R0, #0x00000000`. Other instructions shown include `0x000002DC E59F103C LDR R1, [PC, #0x003C]` and `0x000002E0 F5A10000 STR R0, [R1]`.
- fourth.c:** A window showing the C source code. The code includes comments and assignments for PINSEL0, PINSEL1, and IODIR0, followed by `lcd_init();` and `lcd_string("Hello!");`.
- Command:** A window showing the command `Load "C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 4\Code\`.
- Call Stack + Locals:** A window showing the call stack and local variables.

LCD1  
LM016L

